The Poisson Distribution

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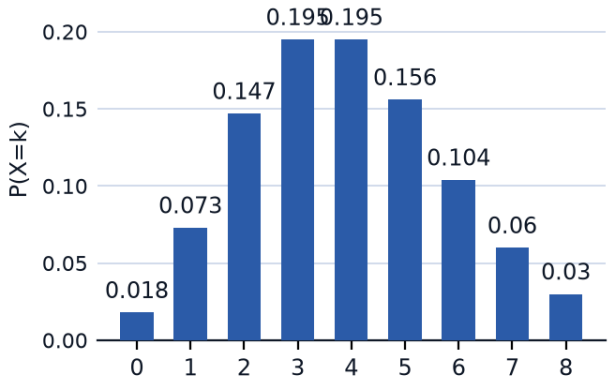
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Conditions for a Poisson distribution**

State the key conditions required for a random variable to be modeled by a Poisson distribution.

2 **[Foundational]** • **Target: Computing Poisson probabilities**

A call center receives an average of 4 calls per minute, modeled by a Poisson distribution with $\lambda = 4$. Find $P(X = 6)$, using $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$, to three decimal places.



3 **[Foundational]** • **Target: Computing Poisson probabilities**

Using the same call center ($\lambda = 4$ calls per minute), find the probability of receiving at most 2 calls in a given minute.

4 **[Foundational]** • **Target: Mean and variance of a Poisson distribution**

State the mean and variance of a Poisson distribution with parameter λ, and explain what is distinctive about this relationship compared to other common distributions.

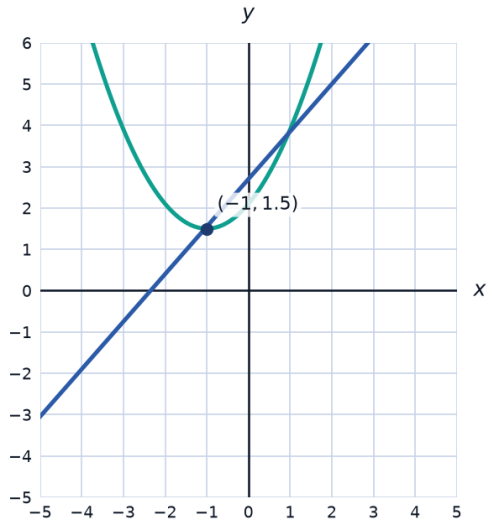
5 **[Foundational]** • **Target: Sums of independent Poisson variables**

Two independent Poisson processes have rates $\lambda_1 = 3$ and $\lambda_2 = 5$ events per hour. If $X_1 \sim \text{Po}(3)$ and $X_2 \sim \text{Po}(5)$, state the distribution of $X_1 + X_2$, and find its mean and variance.

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6 [Foundational] • Target: Poisson as an approximation to the binomial

Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the appropriate value of λ to use.



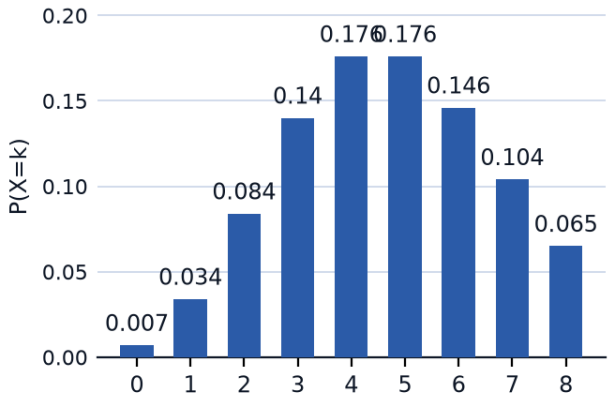
7 [Foundational] • Target: Poisson as an approximation to the binomial

A factory produces 2000 items, each independently defective with probability 0.002. Using the Poisson approximation, find the probability that exactly 3 items are defective, to three decimal places.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Using cumulative Poisson probabilities

A shop receives customers at an average rate of $\lambda = 5$ per 10-minute interval. Find the probability of receiving more than 2 customers in a given 10-minute interval, given $P(X = 0) \approx 0.0067$, $P(X = 1) \approx 0.0337$, and $P(X = 2) \approx 0.0842$.



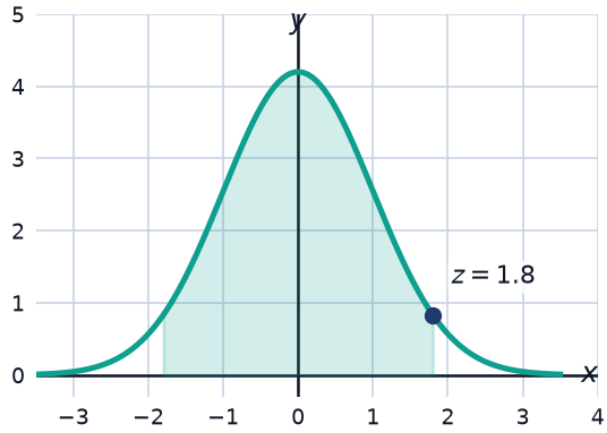
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9 **[Intermediate]** • **Target: Finding an unknown parameter from a given probability**

A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 0) = 0.1353$. Find λ, to the nearest whole number, using $P(X = 0) = e^{-\lambda}$.

10 **[Intermediate]** • **Target: Variance-based reasoning with the Poisson distribution**

A random variable $X \sim \text{Po}(\lambda)$ has standard deviation 3. Find λ, and then find $P(X = \lambda)$, to three decimal places.



11 **[Intermediate]** • **Target: Combining Poisson reasoning with real-world decisions**

A website server crashes on average $\lambda = 0.5$ times per week. The IT team wants to know the probability of at least one crash in a given week, given $P(X = 0) \approx 0.6065$. Should the team be concerned if their target is to keep this probability below 30%?

12 **[Intermediate]** • **Target: Combining a Poisson approximation with conditional probability**

Machine faults occur at an average rate of $\lambda = 1.5$ per day. Find the probability that exactly one fault occurs today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.2231$ and $P(X = 1) \approx 0.3347$.

13 **[Intermediate]** • **Target: Combining Poisson processes over different time intervals**

A radioactive source emits particles at a rate of $\lambda = 2$ per second, modeled by a Poisson distribution. Find $P(X = 3)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.

14 **[Intermediate]** • **Target: Combining Poisson processes over different time intervals**

Explain why the Poisson distribution is not an appropriate model for the number of heads obtained when flipping a fair coin 10 times, referencing the conditions required for a Poisson model.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

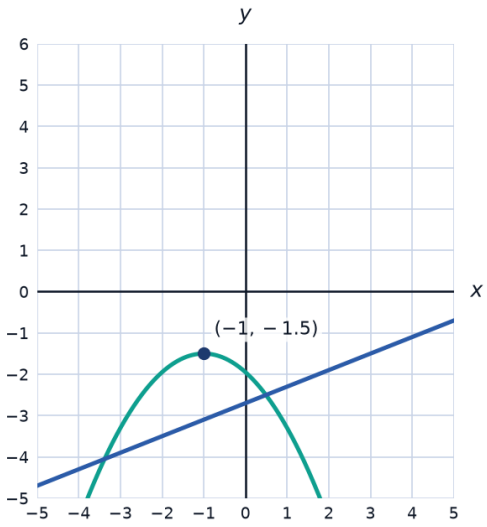
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15 [Challenging] • Target: Comparing two independent Poisson processes • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Two independent Poisson processes have rates $\lambda_A = 3$ and $\lambda_B = 2$ per hour. Find the probability that, in a given hour, process A produces exactly 2 events and process B produces exactly 1 event, to three decimal places, given $P(X_A = 2) \approx 0.2240$ and $P(X_B = 1) \approx 0.2707$.

16 [Challenging] • Target: Rescaling the Poisson parameter for a different interval • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

A radio station receives an average of 12 requests per hour. Find the appropriate Poisson parameter λ to use for a 15-minute segment.



17 [Challenging] • Target: The mode of a Poisson distribution • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

A random variable $X \sim \text{Po}(4)$ has an integer parameter $\lambda = 4$. State the mode(s) of X (the most likely value or values), and explain why a Poisson distribution with an integer λ has two modes, using the ratio $P(X = k)/P(X = k - 1) = \lambda/k$.

18 [Challenging] • Target: Assessing Poisson plausibility from sample data • Cross-Lesson: 5.01 Further Integration Techniques

A biologist counts the number of bacteria colonies in 10 equal-sized petri dishes: $3, 5, 4, 6, 5, 4, 3, 5, 6, 4$. Find the sample mean and the unbiased sample variance, and comment on whether a Poisson model seems plausible for this data.

19 [Challenging] • Target: Mean and variance of a Poisson distribution • Cross-Lesson: 5.02 Differential Equations

The number of accidents on a highway follows a Poisson distribution with mean 1.6 per week. (a) Find the probability of at least 3 accidents in a 2-week period. (b) Given that at least one accident occurs in a week, find the probability that there are exactly 2 accidents in that week.

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20 [Challenging] • Target: Sums of independent Poisson variables • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

In a manufacturing batch, 5% of components are defective. A sample of 12 components is inspected. Find the probability that at least 2 components are defective.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Identifying an appropriate Poisson scenario

Which of the following scenarios is best modeled by a Poisson distribution?

a. The number of typing errors on a randomly selected page of a manuscript.

b. The number of heads in 20 coin tosses.

c. The height of a randomly selected adult.

d. The number of correct answers on a 10-question multiple-choice test, each with 4 options.

22 [Foundational] • Target: Computing Poisson probabilities

Two independent random samples, each of size n, are taken from a normal population with mean μ and known standard deviation σ. From each sample, a symmetric 90% confidence interval for μ is constructed. Given that $\Phi(1.645) = 0.9500$ and $\Phi(2.326) = 0.9900$, where Φ denotes the standard normal cumulative distribution function, find the probability that these two confidence intervals overlap.

a. 0.980

b. 0.900

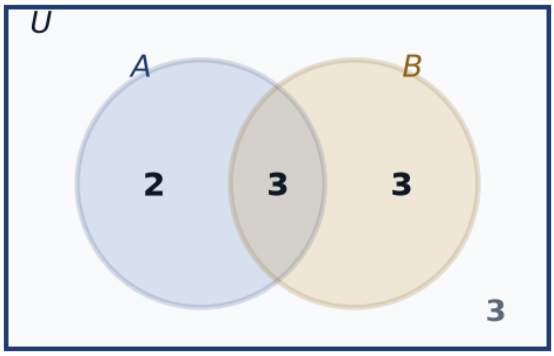
c. 0.810

d. 0.990

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23 [Foundational] • Target: Mean and variance of a Poisson distribution

Flaws in an optical fiber occur according to a Poisson process with a constant mean rate of λ flaws per kilometer. An automated quality-assurance test inspects two independent sections of fiber. The first section has length 1 km and the second section has length 2 km. Let X and Y denote the number of flaws observed in the first and second sections, respectively. Given that the total number of flaws observed across both sections is $X+Y=3$, find the conditional probability $\mathrm{P}(X \geq 1 \mid X+Y=3)$.



a. $\frac{19}{27}$

b. $\frac{8}{27}$

c. $\frac{7}{8}$

d. $\frac{2}{3}$

24 [Foundational] • Target: Sums of independent Poisson variables

A quality control protocol uses a two-stage hypothesis test to monitor the rate of flaws, λ per metre, in the production of high-grade optical fibre. The number of flaws in a length of fibre follows a Poisson distribution. The manufacturer tests: $H_0 : \lambda = 1 \quad \text{against} \quad H_1 : \lambda > 1.$ The two-stage procedure operates as follows: - **Stage 1:** A $1-\text{metre}$ sample is inspected, and the number of flaws X_1 is recorded. If $X_1 \geq 3$, H_0 is rejected immediately. If $X_1 = 0$, H_0 is accepted immediately. If $X_1 \in \{1, 2\}$, the test proceeds to Stage 2. - **Stage 2:** A second, independent $1-\text{metre}$ sample is inspected, and the number of flaws X_2 is recorded. If $X_1 + X_2 \geq 4$, H_0 is rejected; otherwise, H_0 is accepted. Given that the true flaw rate is $\lambda = 2$ per metre, find the probability that a Type II error is committed.

a. $\mathrm{e}^{-2} + 16\mathrm{e}^{-4} \approx 0.428$

b. $16\mathrm{e}^{-4} \approx 0.293$

c. $4\mathrm{e}^{-2} \approx 0.541$

d. $1 - \mathrm{e}^{-1} - \frac{7}{2}\mathrm{e}^{-2} \approx 0.158$

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25 [Foundational] • Target: Poisson as an approximation to the binomial

A factory produces light bulbs. The probability that a randomly selected light bulb is defective is p. The number of defective light bulbs in a random sample of n bulbs follows a binomial distribution $X \sim B(n, p)$. It is known that the mean number of defective bulbs in a sample of $n = 25$ is 2. \\ (i) Calculate the value of p. \\ (ii) For a sample of $n = 50$ bulbs, calculate the probability of finding exactly k defective bulbs, where k is the smallest integer such that $P(X \leq k) > 0.9$. \\ (iii) If a sample of $n = 50$ bulbs is taken, find the smallest value of n' such that the probability of observing at least one defective bulb is greater than 0.999 .

a. 0.04

b. 0.08

c. 0.12

d. 0.16

26 [Foundational] • Target: Using cumulative Poisson probabilities

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$. The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$. Using the standard normal critical value $z_{0.05} = 1.645$: 1. Find the value of σ. 2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.

a. 44

b. 43

c. 11

d. 22

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27 [Foundational] • Target: Finding an unknown parameter from a given probability

A quality control process monitors the weight, W grams, of chocolate bars produced by a machine. A random sample of 50 chocolate bars was taken, and their weights were recorded. The sample mean weight was 102.5 grams, and the sample standard deviation was 3.5 grams. A) Calculate the standard error of the mean. B) Construct a 95% confidence interval for the mean weight of the chocolate bars. C) The machine is calibrated to produce chocolate bars with a mean weight of $\mu_0 = 100$ grams. Based on your confidence interval, what conclusion can be drawn about the calibration of the machine?

a. The machine is likely calibrated correctly, as μ_0 is within the interval.

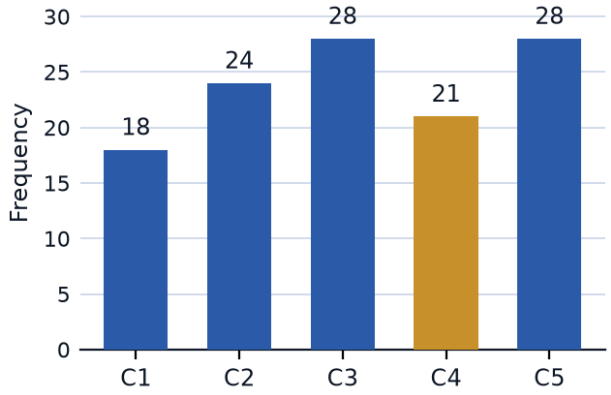
b. The machine is likely not calibrated correctly, as μ_0 is outside the interval.

c. The machine is likely calibrated correctly, as the sample mean is close to μ_0.

d. The machine is likely not calibrated correctly, as the sample standard deviation is too large.

28 [Foundational] • Target: Variance-based reasoning with the Poisson distribution

A research team is studying the impact of different fertilizer types on crop yield. They conducted an experiment with two fertilizer types, A and B, and recorded the yield in kilograms per plot. The data is summarized in the two-way table below. The total number of plots is 200. | Fertilizer Type | Yield < 50 kg | Yield ≥ 50 kg | Total | |---|---|---|---| | A | 30 | 70 | 100 | | B | 50 | 50 | 100 | | Total | 80 | 120 | 200 | Given that a plot was selected at random and its yield was less than 50 kg, what is the probability that it received fertilizer type B? Provide your answer as a fraction in its simplest form.



a. 1/2

b. 5/8

c. 3/8

d. 1/4

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Combining Poisson reasoning with real-world decisions**

A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is $p = \frac{1}{6}$. The number of trials is $n = 4$. This scenario can be modelled by a binomial distribution, denoted as $X \sim B(n, p)$. Which of the following represents the correct parameters for this binomial distribution?

a. $B(4, 1/6)$

b. $B(6, 1/4)$

c. $B(1, 4/6)$

d. $B(4, 4/6)$

30 **[Intermediate]** • **Target: Combining a Poisson approximation with conditional probability**

A factory tests 30 independent components, each with a fixed probability of being defective; X is the number of defective components. Is X binomially distributed?

a. Yes — fixed, independent trials with constant success probability

b. No — the number of trials is not fixed

c. No — trials are not independent / probability is not constant

d. Yes — but only if $n > 30$

31 **[Intermediate]** • **Target: Combining Poisson processes over different time intervals**

A grouped frequency table has class intervals of width 10 starting at 12, with frequencies $[10, 4, 5, 6]$ respectively. Using midpoints, find the mean.

a. 149/5

b. 144/5

c. 154/5

d. 159/5

32 **[Intermediate]** • **Target: Comparing two independent Poisson processes**

$X \sim Po(3)$ and $Y \sim Po(3)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 9

b. 0

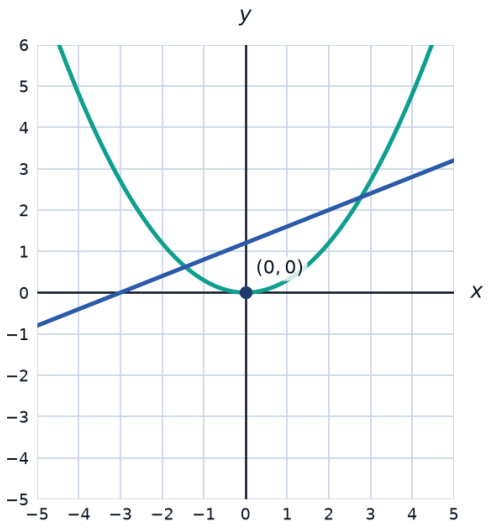
c. 6

d. 7

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33 **[Intermediate]** • **Target: Rescaling the Poisson parameter for a different interval**

A binomial random variable has $n = 200$, $p = \frac{1}{50}$, and is approximated by a Poisson distribution. Find the parameter λ of the approximating Poisson distribution.



a. 50

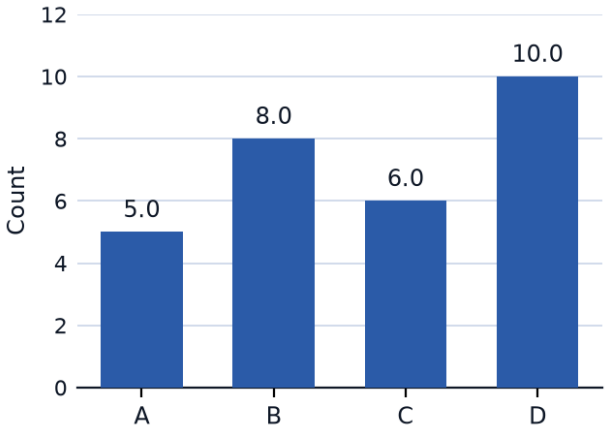
b. 4

c. 200

d. 5

34 **[Intermediate]** • **Target: The mode of a Poisson distribution**

The histogram below shows the distribution of heights of a group of students. The total number of students in the group is 50. If the frequency density for the class interval $60 \leq h < 70$ is 1.2, find the value of k.



a. 10

b. 12

c. 15

d. 20

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35 [Intermediate] • Target: Identifying an appropriate Poisson scenario

A discrete random variable has $P(X = 1) = k$, $P(X = 2) = 2k$, $P(X = 3) = 3k$. Find k.

a. $\frac{5}{12}$

b. $\frac{2}{3}$

c. $\frac{1}{6}$

d. $\frac{5}{6}$

36 [Intermediate] • Target: Assessing Poisson plausibility from sample data

A random variable X equals 2 with probability $\frac{1}{5}$ and 4 with probability $\frac{4}{5}$. Find $\mathrm{Var}(X)$.

a. $\frac{2}{5}$

b. $\frac{16}{25}$

c. $\frac{33}{25}$

d. $\frac{39}{50}$

37 [Intermediate] • Target: Conditions for a Poisson distribution

If $X \sim \mathrm{Po}(4)$, find $P(X = 3)$, correct to 4 decimal places.

a. 0.2454

b. 0.1563

c. 0.1465

d. 0.1954

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Computing Poisson probabilities • Cross-Lesson: 5.01 Further Integration Techniques

Calls arrive at a technical support desk according to a Poisson process with an average rate of λ calls per hour. It is known that the probability of receiving exactly 1 call in a given hour is 1.5 times the probability of receiving exactly 3 calls in that same hour. Find the probability that the desk receives exactly 4 calls during a randomly chosen 3-hour period.

a. $54\mathrm{e}^{-6} \approx 0.134$

b. $\frac{2}{3}\mathrm{e}^{-2} \approx 0.0902$

c. $\frac{2187}{8}\mathrm{e}^{-9} \approx 0.0337$

d. $\frac{625}{24}\mathrm{e}^{-5} \approx 0.175$

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39 [Challenging] • Target: Mean and variance of a Poisson distribution • Cross-Lesson: 5.02 Differential Equations

The scores on a standardized test are normally distributed with a mean of 100 and a standard deviation of 15. A new scoring algorithm is implemented, and the scores are transformed using the rule $Y = aX + b$, where X is the original score and Y is the new score. The new distribution of scores, Y, also follows a normal distribution with a mean of 120 and a standard deviation of 22.5. Calculate the value of a and the value of b.

a. a = 1.5, b = -30

b. a = -1.5, b = 270

c. a = 1.5, b = 30

d. a = 0.67, b = 53

40 [Challenging] • Target: Sums of independent Poisson variables • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

The unique real root α of the equation $x^3 + 2x - k = 0,$ where k is a positive constant, is estimated using the iterative formula $x_{n+1} = \frac{k}{x_n^2 + 2}$. The sequence of iterations converges locally to α (for an initial approximation x_0 sufficiently close to α) if and only if $|g'(\alpha)| < 1$, where $g(x) = \frac{k}{x^2 + 2}$. Determine the complete set of values of k for which this iteration converges locally to α.

a. $0 < k < 4\sqrt{2}$

b. $0 < k < 2\sqrt{2}$

c. $0 < k < 3$

d. $0 < k < 12$

41 [Challenging] • Target: Poisson as an approximation to the binomial • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A fair coin is tossed 3 times. Let X be the number of heads obtained. What is the probability that X = 2?

a. $\frac{1}{8}$

b. $\frac{3}{8}$

c. $\frac{1}{2}$

d. $\frac{7}{8}$

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42 [Challenging] • Target: Using cumulative Poisson probabilities • Cross-Lesson: 6.01 Matrices (AI HL)

A machine produces metal rods whose lengths are normally distributed with unknown mean μ cm and known standard deviation $\sigma = 0.8$ cm. A quality control manager tests the hypotheses $H_0 : \mu = 25.0 \quad \text{against} \quad H_1 : \mu < 25.0$ at the 5% level of significance. A random sample of 16 rods yields a sample mean length of $\bar{x} = 24.6$ cm. Given that the critical value for this test is $z_{\text{crit}} = -1.645$, which of the following gives the correct test statistic z and the corresponding conclusion?

a. $z = -2.0$; reject H_0 because $-2.0 < -1.645$

b. $z = -2.0$; do not reject H_0 because $-2.0 < -1.645$

c. $z = -0.5$; do not reject H_0 because $-0.5 > -1.645$

d. $z = -8.0$; reject H_0 because $-8.0 < -1.645$

43 [Challenging] • Target: Finding an unknown parameter from a given probability • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

The weights of bags of sugar are normally distributed with a mean of 1000 g and a standard deviation of 5 g. Let X be the weight of a randomly selected bag of sugar. Calculate the probability that a bag of sugar weighs between 990 g and 1010 g.

a. 0.9545

b. 0.6827

c. 0.9772

d. 0.0228

44 [Challenging] • Target: Variance-based reasoning with the Poisson distribution • Cross-Lesson: 6.03 Markov Chains (AI HL)

The random variable X has a Poisson distribution with parameter λ, where $\lambda > 0$. Given that $\mathrm{P}(X = 3) = \mathrm{P}(X = 4)$, find the value of λ.

a. 4

b. 3

c. 12

d. $\frac{4}{3}$

45 [Challenging] • Target: Combining Poisson reasoning with real-world decisions • Cross-Lesson: 1.01 Proof by Mathematical Induction

The random variable X has a Poisson distribution with mean $\lambda > 0$. Given that $3\,\mathrm{P}(X = 2) = \mathrm{P}(X = 4)$, find the value of λ.

a. 6

b. 36

c. 12

d. $\sqrt{6}$

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